Notes

1 Langevin Dynamics Integrator: BAOAB method

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In this work, we simulated the Langevin Dynamics by the BAOAB method. This method was implemented by B. Leimkuhler and C. Matthews [2]. The aim of their work is to compare Langevin dynamics integrators in term of their invariant distributions and demonstrate a superconvergence property of a certain method in the high friction limit (or in the null mass limit).

The methods that are considered are splitting integrators that decompose the stochastic vector field of Langevin Dynamics into simpler vector fields which can be solved exactly. Nevertheless, the composition method that results cannot be directly related to a stochastic differential equation, but nonetheless the invariant measure associated to the numerical method can be derived, using the Baker-Campbell-Hausdorff (BCH) expansion, as an asymptotic series in two parameters: the stepsize and the coefficient γ^{-1}. The authors stress that their approach focuses on the truncation error of the invariant distribution, thus the greatest benefit would be seen only when the statistical error is well controlled. Furthermore, they observe that the least biased scheme from the point of wiew of the error introduced in configurational sampling is also as efficient as the alternatives and the most robust with respect the variation of parameters (stepsize and γ^{-1}), thus they emphasise that there is no price for the improvement in accuracy.

Numerically they implemented the various methods considered and compared the accuracy of configurational sampling for different values of the friction parameter γ and a range of timesteps.

Their results show that the best method to sampling of the configurational Gibbs–Boltzmann distribution[1] is the BAOAB method and in the context of molecular dynamics it is a cheap and easy to implement scheme using only a single force vector per timestep:

$$\begin{cases} p_{n+1/2} = p_n - \frac{1}{2}\,\nabla U\left(x_n\right)\,dt \\ x_{n+1/2} = x_n + \frac{1}{2}\,M^{-1}\,p_{n+1/2}\,dt \\ \hat{p}_{n+1/2} = c_1\,p_{n+1/2} + c_2\,M^{1/2}\,R_{n+1} \\ x_{n+1} = x_{n+1/2} + \frac{1}{2}\,M^{-1}\,\hat{p}_{n+1/2}\,dt \\ p_{n+1} = \hat{p}_{n+1/2} - \frac{1}{2}\,\nabla U\left(x_{n+1}\right)\,dt \end{cases}$$

where R_i is an N-vector of iid normal random numbers with mean 0 and variance 1, M is the diagonal mass matrix, dt is the timestep,

$$c_1 = e^{-\gamma\,dt} \quad \text{and} \quad c_2 = \sqrt{\left(1 - c_1^2\right)/\beta}$$

are evaluated for a given friction parameter γ and a given temperature parameter $\beta = (K_B T)^{-1}$, where K_B is the Boltzmann's constant.

[1] This means the calculation of functions averages of phase space variables of a molecular N-body system modeled at constant temperature.

1.1 An idea of how the BAOAB method was derived

1.1.1 Background

An Itô stochastic differential equation (SDE) can be written as

$$d\mathbf{x}(t) = \mathbf{a}\,[\mathbf{x}(t), t]\,dt + \mathbf{b}\,[\mathbf{x}(t), t]\,d\mathbf{W}(t)\,,$$

where $\mathbf{x}$ are the n stochastic process, $\mathbf{a}(\mathbf{x}, t)$ and $\mathbf{b}(\mathbf{x}, t)$ are certain know functions,

$$d\mathbf{W} \equiv \mathbf{W}(t + dt) - \mathbf{W}(t) = \boldsymbol{\xi}(t)\,dt$$

is an n variable Wiener process and $\boldsymbol{\xi}(t)$ is the white noise.
This SDE is equivalent an Fokker-Planck equation if the stochastic process described by a conditional probability density $p(\mathbf{x}, t\,|\,\mathbf{x_0}, t_0) \equiv p$ satisfy

$$\partial_t\, p = \mathscr{L}(\mathbf{x}, t)\, p = -\sum_i \partial_i\,[A_i(\mathbf{x}, t)\, p\,] + \frac{1}{2}\sum_{ij} \partial_i\partial_j\,[B_{ij}(\mathbf{x}, t)\, p\,]$$

where $\mathscr{L}(\mathbf{x}, t)$ is a second-order differential operator and the coefficients of RHS are such that $\mathbf{A}(\mathbf{x}, t) = \mathbf{a}(\mathbf{x}, t)$ and $\underline{\mathbf{B}}(\mathbf{x}, t) = \mathbf{b}(\mathbf{x}, t)\mathbf{b}^T(\mathbf{x}, t)$. The Fokker-Planck equation describes diffusion processes, the vector $\mathbf{A}(\mathbf{x}, t)$ is know as drift vector and the matrix $\underline{\mathbf{B}}(\mathbf{x}, t)$ is know as the diffusion matrix[2].

Let us consider an Orstein-Uhlenbeck (OU) SDE, that is, an Itô equation where the coefficients are

$$\mathbf{a}(\mathbf{x}) = -k\,\mathbf{x} \quad \text{and} \quad b = \sqrt{D}$$

where k and D are positive parameters. We can easily show in the stationary limit ($t \gg t_0$) that the diffusive parameter D depends on parameters k and T: $D = (2k)/\beta$. The idea is to compute average and variance of the solution of the OU-SDE in the stationary limit and then apply the equipartition theorem[3].

1.1.2 Langevin Dynamics

Langevin dynamics combines the OU stochastic vector field with a conservative dynamics, so we are assuming that the stochastic force is not affected by the potential:

$$M\frac{d^2\mathbf{x}}{dt^2} = -\boldsymbol{\nabla} U(\mathbf{x}) - \gamma\,\frac{d\mathbf{x}}{dt} + \sqrt{(2\gamma)/\beta}\,\boldsymbol{\xi}(t)$$

[2] The diffusion matrix is positive semidefinite and symmetric as a result of its definition [1].

[3] The solution of one dimensional OU-SDE can be computed by the Itô's formula applied to $y = xe^{k\,t}$[1]. We get

$$x(t) = x_0(t_0)\,e^{-k(t-t_0)} + \sqrt{D}\int_{t_0}^{t} dW(t')e^{-k(t-t')}\,,$$

and the average and variance are respectively:

$$\langle x(t)\rangle = \langle x_0(t_0)\rangle\; e^{-k(t-t_0)}\,,$$
$$\Big\langle (x(t) - \langle x(t)\rangle)^2 \Big\rangle = \left[Var(x_0) - \frac{D}{2k}\right] e^{-2k(t-t_0)} + \frac{D}{2k}\,.$$

where we introduced physical quantities: $\mathbf{x}$ is the canonical coordinate of position, γ is the friction parameter and $\beta = (K_B T)^{-1}$ is the temperature parameter. We can rewrite the equation as

$$\begin{cases} d\mathbf{x} = \mathbf{v}\, dt \\ M d\mathbf{v} = [-\boldsymbol{\nabla} U(\mathbf{x}) - \gamma\, \mathbf{v}]\ dt + \sqrt{(2\gamma)/\beta}\, d\mathbf{W} \end{cases}$$

and for the canonical coordinates we get

$$\begin{cases} d\mathbf{x} = M^{-1}\, \mathbf{p}\, dt \\ d\mathbf{p} = [-\boldsymbol{\nabla} U(\mathbf{x}) - \tilde{\gamma}\, \mathbf{p}]\ dt + \sqrt{(2\tilde{\gamma})/\beta}\, M^{1/2} d\mathbf{W} \end{cases}$$

where $\tilde{\gamma} = \gamma/M$. From the system of two first-order equations can we compute the drift and diffusion coefficients

$$\begin{aligned} (A_{\mathbf{x}})_i &= M^{-1}\, p_i\,; & (B_{\mathbf{xx}})_{ij} &= (B_{\mathbf{xp}})_{ij} = 0\,; \\ (A_{\mathbf{p}})_i &= (-\boldsymbol{\nabla} U(\mathbf{x}) - \tilde{\gamma}\, \mathbf{p})_i\ ; & (B_{\mathbf{pp}})_{ij} &= \frac{2\tilde{\gamma}}{\beta}\, M\, \delta_{ij}\,; \end{aligned}$$

and thus determine the Fokker-Planck operator for the Langevin dynamics

$$\begin{aligned} \mathscr{L}_{LD}\, \varphi = &-M^{-1} \sum_i p_i\, \partial_{x_i}\, \varphi\ +\ \boldsymbol{\nabla} U(\mathbf{x}) \sum_i \partial_{p_i}\, \varphi\ + \\ &+\ \tilde{\gamma} \sum_i \partial_{p_i}\, p_i\, \varphi\ +\ \tilde{\gamma} \sum_i\ p_i\, \partial_{p_i}\, \varphi\ +\ \tilde{\gamma}\, K_B T \sum_i M\, \partial_i^2\, \varphi\ = \\ &= -M^{-1} \mathbf{p} \cdot \boldsymbol{\nabla}_x\, \varphi\ +\ \boldsymbol{\nabla} U(\mathbf{x}) \cdot \boldsymbol{\nabla}_p\, \varphi\ +\ \tilde{\gamma}\, \boldsymbol{\nabla}_p (\mathbf{p}\, \varphi) + \tilde{\gamma}\, K_B T \boldsymbol{\Delta}_p\, \varphi\,, \end{aligned}$$

where $\boldsymbol{\Delta}_p$ is the mass-weighted Laplacian in the momenta and φ is a test function [3].

1.1.3 Splitting methods

References

[1] C. W. Gardiner. *Handbook of Stochastic Methods for Physics, Chemistry and the Natural Sciences*. Springer-Verlag Berlin Heidelberg, 1985.

[2] B. Leimkuhler, C. Matthews. Rational construction of stochastic numerical methods for molecular sampling. *Applied Mathematics Research eXpress*, 2013(1):34–56, 2012.

[3] H. Risken. *The Fokker-Planck Equation Methods of Solution and Applications*. Springer-Verlag Berlin Heidelberg, 1996.